



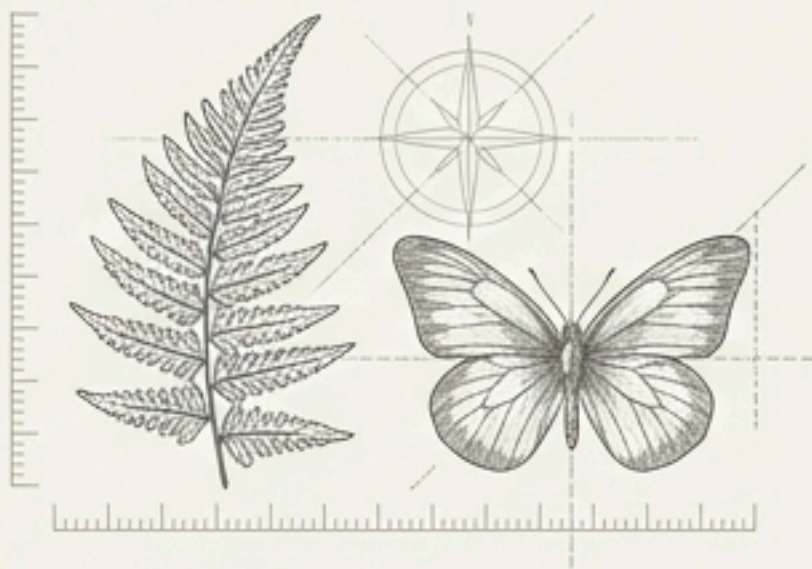
The Living World: From Diversity to Systematics

Understanding the 1.7–1.8 million species that share our planet.

UNIT 1: DIVERSITY IN THE LIVING WORLD

The Magnitude of Biodiversity

Biology is the science of life forms. The range and variety of organisms is vast and grows as we increase our area of observation. This sheer volume refers to biodiversity—the number and types of organisms present on earth.



1.7–1.8
Million Species

Number of known
and described species.



The Tower of Babel: The Problem with Local Names

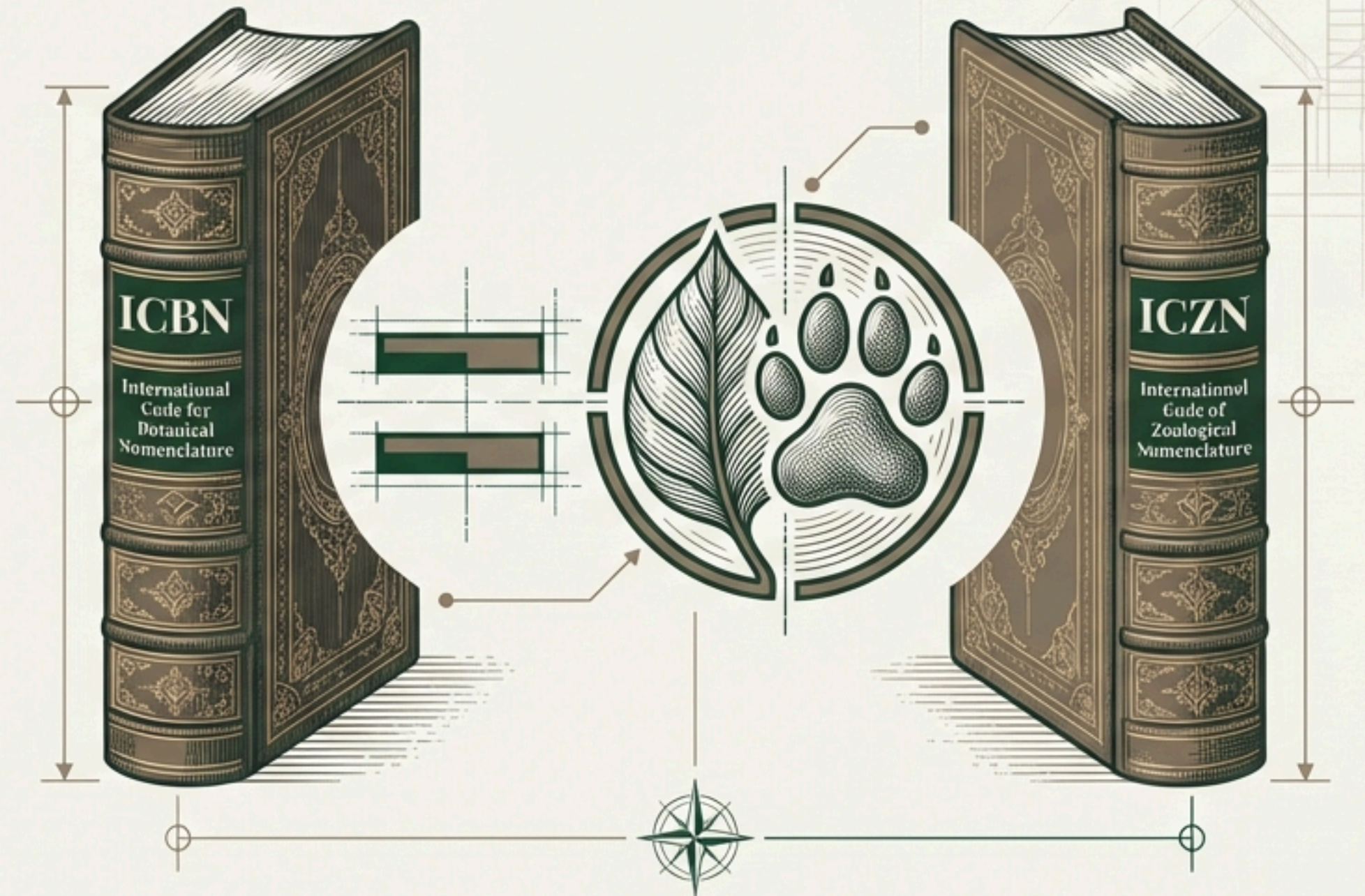
The Conflict: Millions of organisms exist, but local names vary by region and country.

The Consequence: Global scientific communication collapses without a standard system.



Nomenclature: The Universal Language

- **Nomenclature:** Standardising names so one organism is known by the same name globally.
- **Prerequisite:** Identification (Describing the organism correctly).
- **Key Principle:** One Organism = One Scientific Name.



Binomial Nomenclature

The System of Carolus Linnaeus

Mangifera indica Linn.

Generic Name (Genus).
Capitalised.

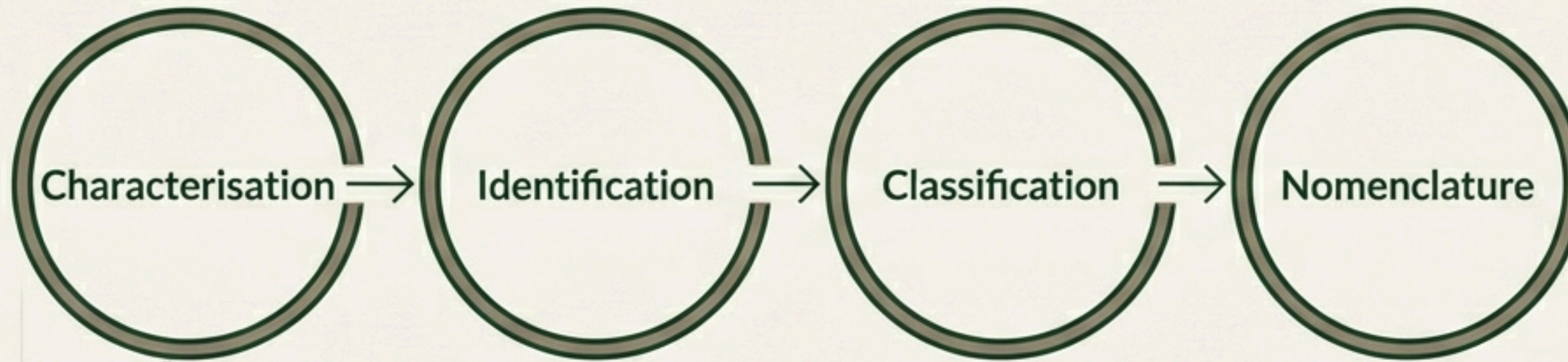
Specific Epithet (Species).
Small letter.

Author Citation.
Abbreviated.

Universal Rules

- | | | | | | | | | | | | | |
|---|--|--|--|--|--|--|--|--|--|--|--|--|
| 1. Latin origin (written in <i>italics</i>). | | | | | | | | | | | | |
| 2. First word is Genus; second is Species. | | | | | | | | | | | | |
| 3. Handwritten names must be underlined separately. | | | | | | | | | | | | |
| 4. Author name appears at the end. | | | | | | | | | | | | |

Taxonomy and Systematics



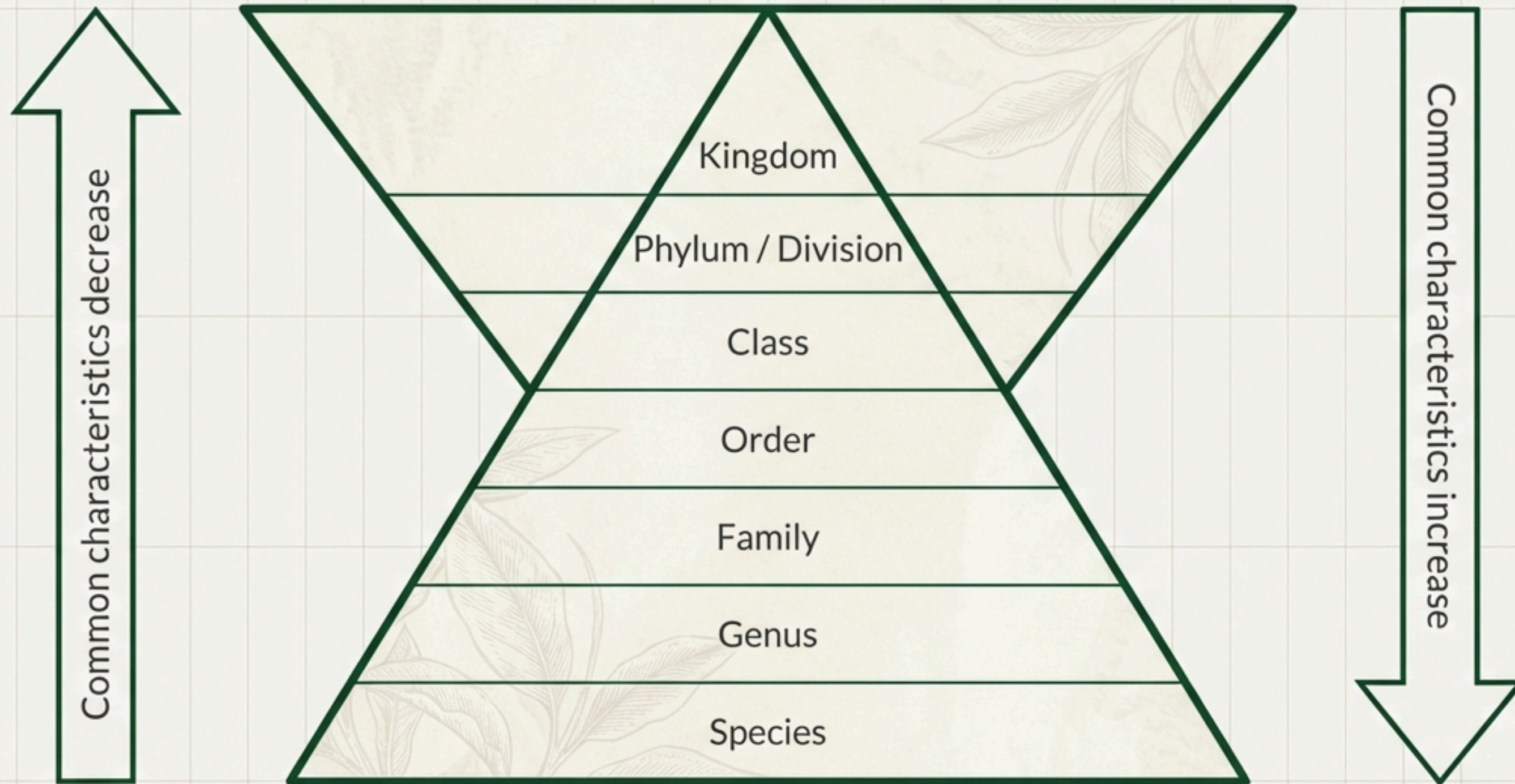
Processes Basic to Taxonomy

Systematics

**Taxonomy +
Evolutionary
Relationships
(Phylogeny)**

Linnaeus used 'Systema Naturae' as his publication title.

The Taxonomic Hierarchy



The Foundation: Species and Genus

Genus: *Solanum*



Species: *tuberosum*



Species: *melongena*

Genus: *Panthera*



Species: *leo*

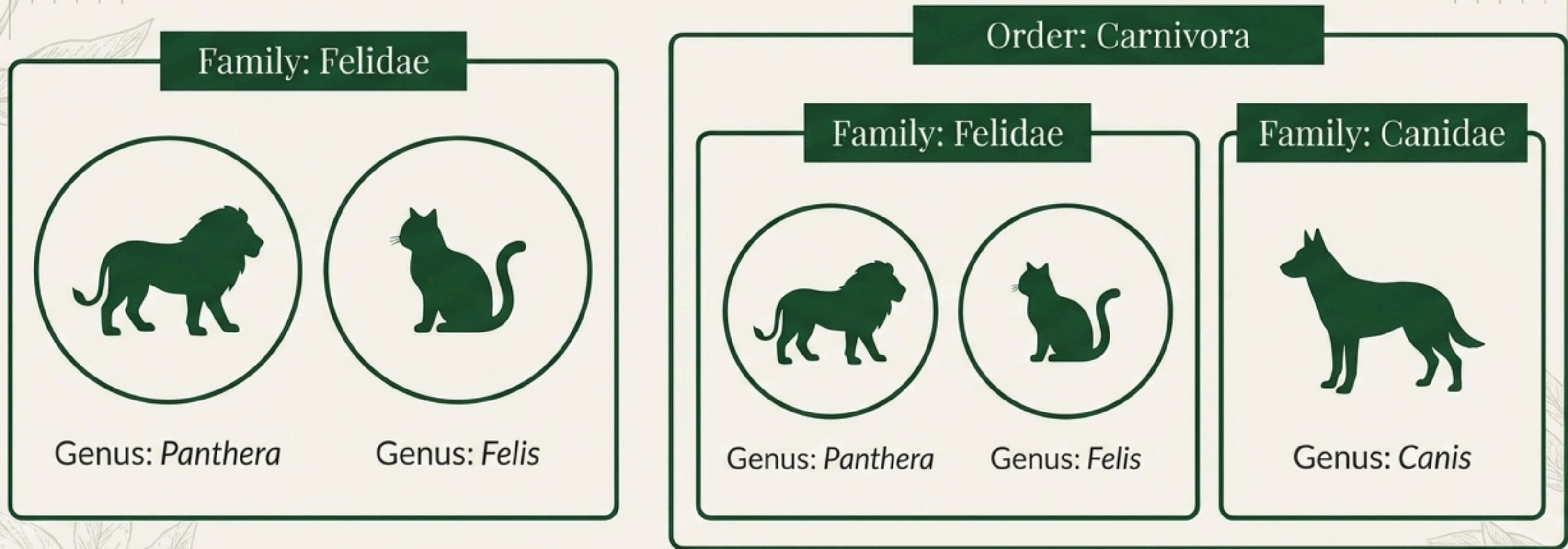


Species: *tigris*

Species: A group with fundamental similarities.

Genus: Aggregates of closely related species.

Aggregating Diversity: Family and Order



Families are groups of related genera.
Orders are assemblages of families exhibiting a few similar characters.

Broad Categories: Class and Phylum

Phylum: Chordata

Notochord, dorsal hollow neural system.



Class: Mammalia

External ears, body hair



Order Primata



Order Carnivora

Phylum is the equivalent of 'Division' in plants.

The Highest Rank: Kingdom



Kingdom Plantae

Comprises all plants from various divisions.



Kingdom Animalia

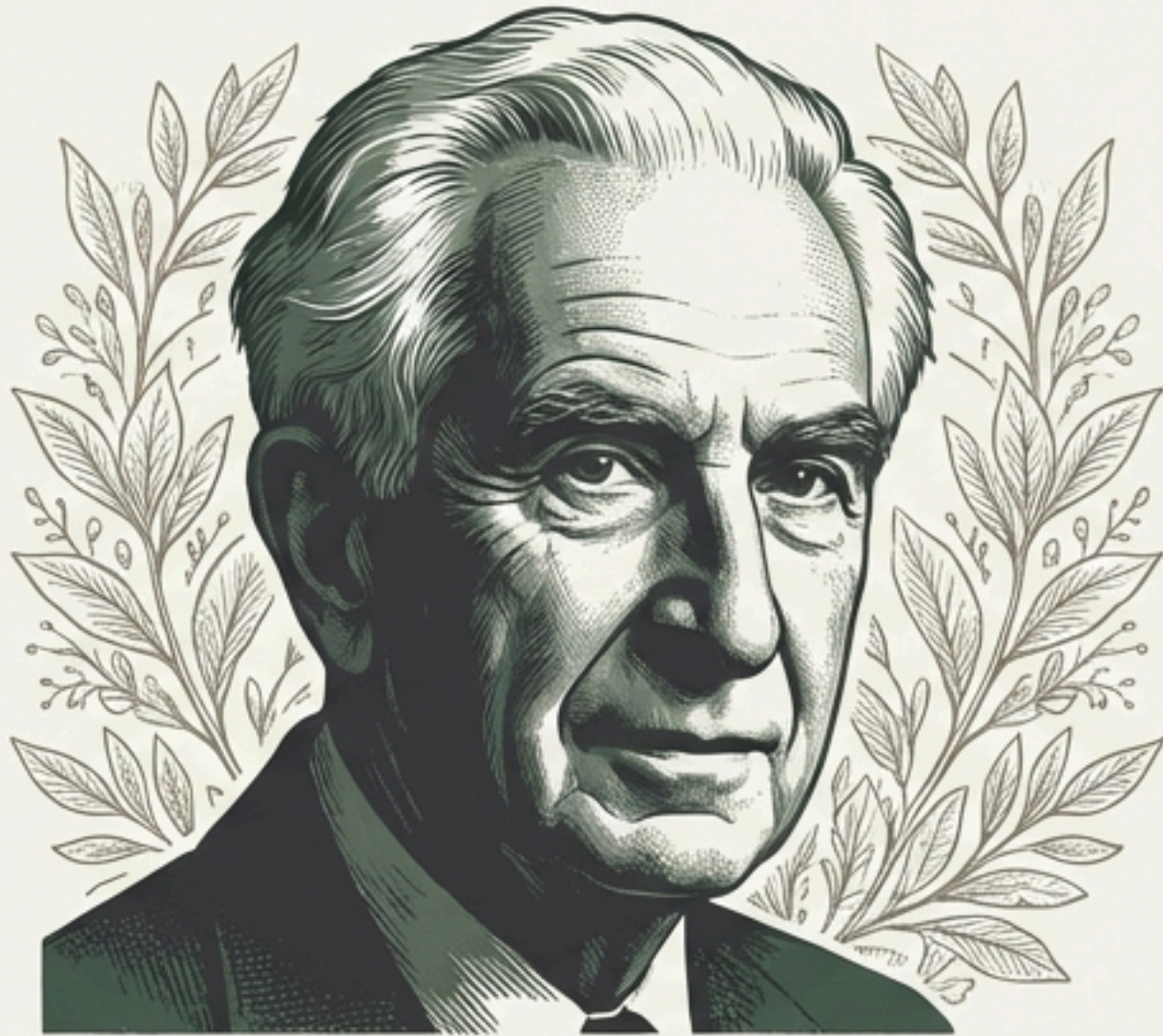
Contains all animals belonging to various phyla.

Comparative Taxonomy

Common Name	Biological Name	Genus	Family	Order	Class	Phylum/ Division
Man	<i>Homo sapiens</i>	<i>Homo</i>	Hominidae	Primata	Mammalia	Chordata
Housefly	<i>Musca domestica</i>	<i>Musca</i>	Muscidae	Diptera	Insecta	Arthropoda
Mango	<i>Mangifera indica</i>	<i>Mangifera</i>	Anacardiaceae	Sapindales	Dicotyledonae	Angiospermae
Wheat	<i>Triticum aestivum</i>	<i>Triticum</i>	Poaceae	Poales	Monocotyledonae	Angiospermae

The Darwin of the 20th Century

Ernst Mayr (1904–2004)

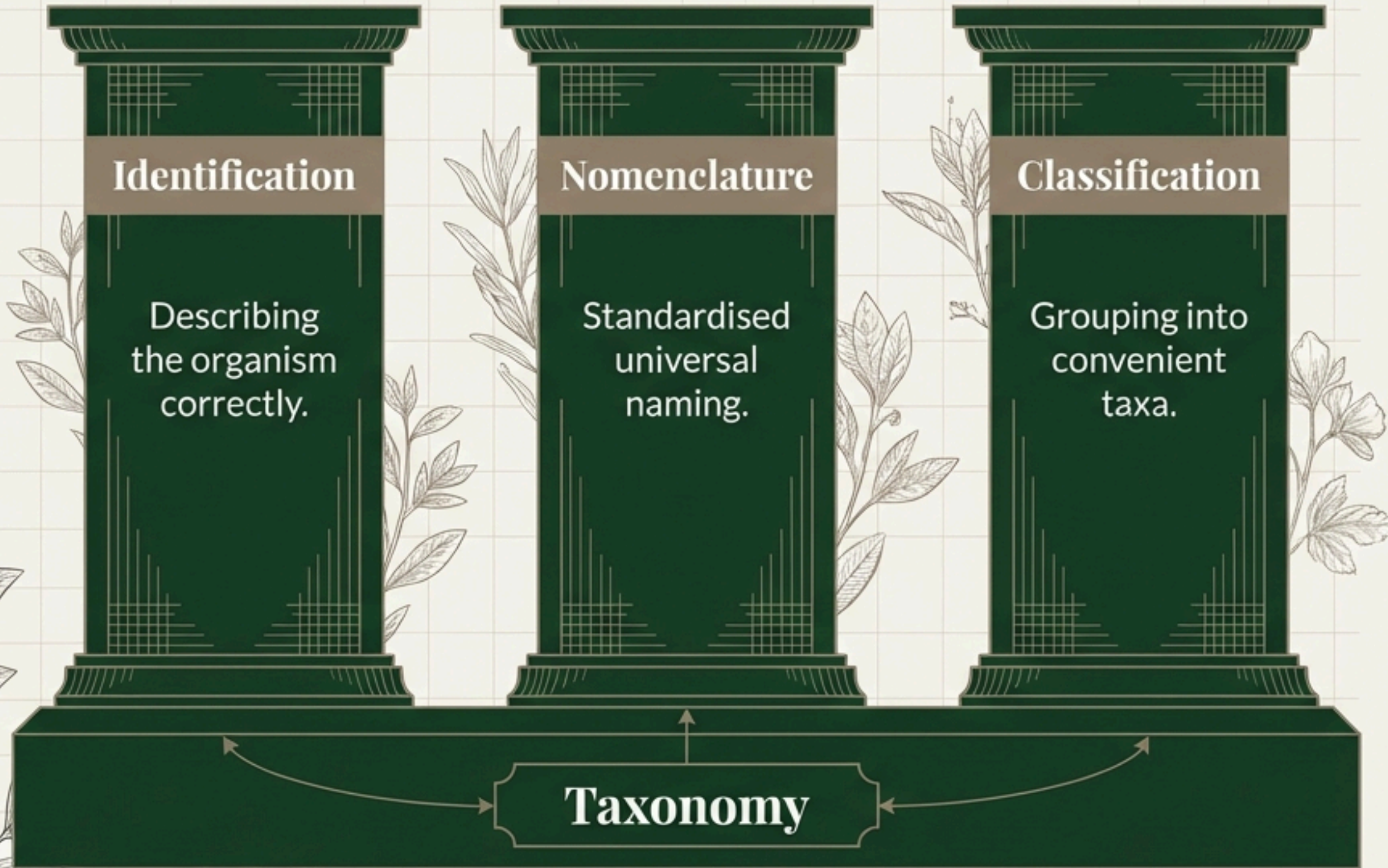


Harvard University evolutionary biologist and one of the 100 greatest scientists of all time.

Key Contribution: He made the origin of species diversity the central question of evolutionary biology and pioneered the definition of a biological species.

Awards: The Triple Crown of Biology (Balzan Prize, International Prize for Biology, Crafoord Prize).

Why We Classify



Taxonomy allows us to understand our bio-resources and navigate the diversity of millions of species. It is the foundation of agriculture, forestry, and industry.

Reflections & Exercises

Test your understanding of the living world.

1. Why are living organisms classified?
2. What does the scientific name '*Mangifera indica* Linn.' indicate about the species' discovery?
3. As we move from Species to Kingdom, what happens to the number of common characteristics?
4. Define a Taxon.